

Capstone Poster

SecureWave Technology

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Introduction

SecureWave Technology operates within a **Microsoft domain environment** utilizing **Windows Server 2019** for network management and file sharing. This setup uses **Active Directory Domain Services (ADDS)** to centrally manage all network computers, users, and resources. However, SecureWave faces significant security challenges due to **irregular system updates**, a **lack of formal patch management**, and **outdated firmware** on critical network devices. These issues create risks like system instability, compliance violations, and a high vulnerability to cyberattacks. This report will detail these vulnerabilities and offer specific, **actionable recommendations** to improve SecureWave's security posture while maintaining continuous operation.

Sporadic Patch Management

SecureWave Technology lacks a structured patch management policy, resulting in **inconsistent and sporadic updates** across its **Microsoft domain environment**. This exposes critical systems, including domain controllers, to severe risks such as exploitation of unpatched vulnerabilities, system instability, and compliance violations.

Figure 1

Top 10 Threat Vulnerabilities



Note. From Top 10 Threat Vulnerabilities. (2025). DeepStrike. Retrieved 2025, (<https://cdn.sanity.io/images/a3jopls3/testdataset/64db5c1b9a1a822d0b419f6e83c35b991c13d843-1536x1024.jpg>).

Sporadic Patch Management

Without routine updates, attackers can leverage known vulnerabilities like CVE-2024-49113, which enables remote code execution and compromises Windows servers. Additionally, connecting directly to Microsoft update servers without using Windows Server Update Services (WSUS) increases the risk of man-in-the-middle attacks if TLS connections are misconfigured. Poor planning and absence of rollback procedures further heighten operational risks, including downtime and broken integrations. Failure to address these issues can lead to regulatory non-compliance under frameworks like GDPR or PIPEDA, financial penalties, and loss of business trust. Implementing a consistent patching strategy and centralized update management is critical to reducing SecureWave's attack surface and maintaining system integrity.

Neglecting Firmware Updates on Network Devices

SecureWave Technology's **routers and switches** are running **outdated firmware**, creating a critical security gap in the network infrastructure. Firmware updates often include essential security patches and protocol enhancements; without them, these devices become prime targets for attackers seeking to reroute traffic, inject malicious commands, or compromise network integrity. Outdated firmware can also lead to compatibility issues with modern devices, degraded performance, and increased risk of network outages.

Figure 2

An Example of a Network Device (Router) that Requires Updates on Firmware



Neglecting Firmware Updates on Network Devices

Furthermore, security tools such as SIEM systems may fail to monitor traffic effectively if older protocols are in use, leaving blind spots for attackers to exploit. Studies indicate that up to 70% of router vulnerabilities stem from obsolete firmware, underscoring the severity of this risk. Beyond security concerns, outdated firmware can cause bottlenecks, unstable connections, and incompatibility with newer standards, impacting overall business operations. To mitigate these risks, SecureWave must implement a routine firmware update schedule, maintain an inventory of device versions, and replace hardware that cannot support current security standards. Consistent firmware management is essential to ensure network resilience and prevent attackers from exploiting these weaknesses.

Figure 3

An Example of a Network Device (Router) that Requires Updates on Firmware



Lack of Change Management and Documentation

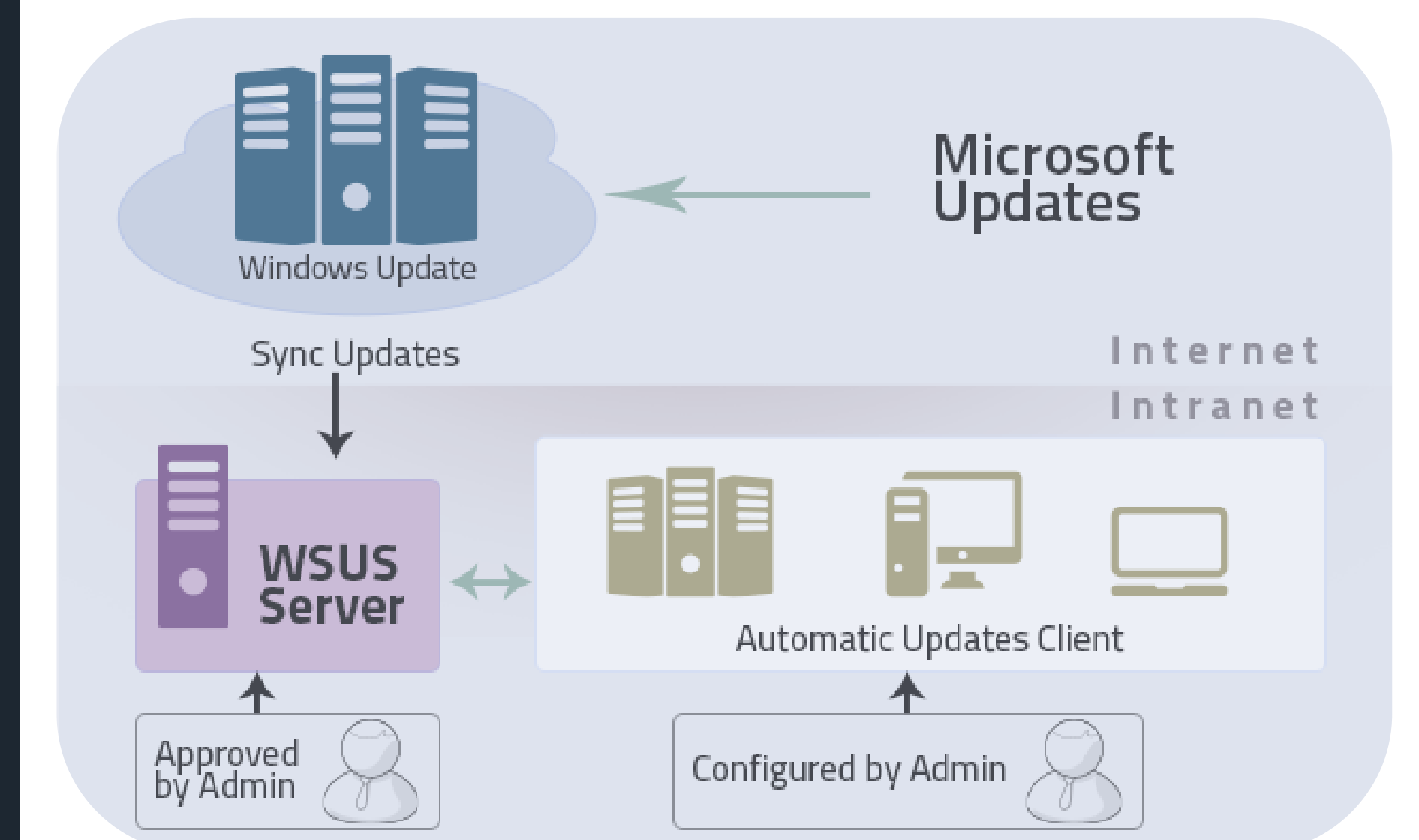
SecureWave Technology does not have a **formal change management process** or **documentation** for **system updates**, creating significant security and operational risks. Without structured procedures, updates are applied inconsistently, leaving critical servers and network devices vulnerable to zero-day exploits and compatibility issues. The absence of centralized control also increases the likelihood of misconfigurations, service interruptions, and exposure to attacks such as man-in-the-middle exploits when connecting directly to Microsoft update servers. Furthermore, undocumented changes make troubleshooting difficult and hinder compliance with regulatory standards, potentially resulting in fines and loss of trust.

Lack of Change Management and Documentation

Implementing a change management framework ensures updates are tested, approved, and deployed systematically, reducing downtime and preventing unauthorized modifications. Tools like Windows Server Update Services (WSUS) can centralize patch deployment, enforce update policies, and maintain detailed logs for auditing purposes. By adopting these practices, SecureWave can improve security, maintain system integrity, and ensure operational continuity across its Microsoft domain environment.

Figure 4

Flow Chart of Patch Management using Windows Server Update Services



Conclusion

SecureWave Technology's current systems and network maintenance practices reveal significant cybersecurity vulnerabilities due to unstructured update policies, dependence on direct Microsoft server connections, and outdated firmware on devices. To minimize these risks, implementing patch management, documented change control, and keeping firmware updated are essential. Moreover, adopting proactive security measures and continuous monitoring will help SecureWave protect its infrastructure and maintain client trust in a high-risk digital environment.